

Cardinalities in finite monoids of G -equivariant functions

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Abstract

A set with a group action is referred to as a G -set, and the set of functions that commute with this action forms a monoid under function composition. This paper examines the case where the G -set is finite, which implies that the monoid of G -equivariant functions is also finite. The document provides formulas for calculating the cardinality of this monoid, its group of units, and explores special cases of G -equivariant functions, known as fixing elementary collapsings. All of these results are expressed in terms of specific properties of the G -set, including the number of orbits and certain indices of the subgroups acting as stabilizers.

Keywords: Group actions, G -sets, G -equivariant function, cardinality.

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1 Introduction

For any group G , a G -set is simply a set X on which G acts; that is, there exists a function $\cdot : G \times X \rightarrow X$ such that $e \cdot x = x$ for all $x \in X$ and $g \cdot (h \cdot x) = (gh) \cdot x$ for all $x \in X, g, h \in G$. In the context of semigroup theory, G -sets are also known as G -acts. A G -equivariant transformation of X , or a G -endomorphism of X , is a function $\tau : X \rightarrow X$ such that $\tau(g \cdot x) = g \cdot \tau(x)$ for all $g \in G, x \in X$. These maps are fundamental in the category of G -sets and find applications in various branches of mathematics such as equivariant topology, representation theory, and statistical inference.

The set of all G -equivariant transformations of X , which are functions that commute with the group action, forms a monoid under function composition. We denote this monoid as $\text{End}_G(X)$, and its group of units, consisting of all bijective G -equivariant transformations, as $\text{Aut}_G(X)$. These objects have been extensively studied in various contexts (see [2], [3], [10], [16]). Many examples of objects with group actions and associated G -equivariant functions have been explored, such as cellular automata, which have been the motivation

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